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FAMOS

A Multi-State Articulation Transformer Reconstructs Joint Parameters from Sparse Unordered Point Clouds Without Category Priors

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Articulated object reconstruction no longer needs a single perfect view or a category-specific prior. *FAMOS* (Feed-forward Articulation Modeling from Sparse Observations) accepts an unordered set of partial point clouds of the same object in different joint states, aggregates articulation evidence across views with a Multi-state Articulation Transformer, and predicts movable-part segmentation plus full joint parameters in a single forward pass—outperforming the best prior feed-forward method by up to 27.8 points on three standard benchmarks.

AT A GLANCE

Problem: Existing feed-forward methods infer articulation from a single observation and rely on strong category priors that fail on novel instances and unusual joint configurations.

Why it matters: Robots that can estimate joint parameters of arbitrary objects from a few casual scans would generalise to unstructured environments without per-object calibration.

Method: Multi-state Articulation Transformer with alternating state-wise and global attention; Observed Articulation Span (OAS) objective to exploit motion diversity; procedural synthetic data generation to overcome dataset scarcity.

Result: +8.1, +27.8, +21.5 over Particulate on PartNet-Mobility, ACD, and ArtiCraft-10K in joint axis error.

Evidence: Evaluation on three established articulated-object benchmarks with ablations over attention variants, OAS, and observation count.

The Big Picture

Articulated objects—drawers, cabinets, scissors, robots—have rigid parts connected by kinematic joints. Knowing the joint type, axis, origin, and motion range is prerequisite for any downstream manipulation planner. The classical approach requires dense RGBD scans or repeated observations across many configurations; modern feed-

forward methods aim to infer joint parameters in a single forward pass but have so far been limited to one observation and one category at a time.

FAMOS abandons both restrictions. Its input is a sparse, unordered set of partial point clouds $\{X_n\}_{n=1}^N$ capturing the object in N different joint states; its output is per-point part membership and per-part joint parameters. Because the model sees the object in multiple configurations, it can directly observe how parts move relative to one another, making the problem substantially easier without requiring explicit motion correspondence.

Why Existing Approaches Fall Short

Optimization-based methods (A-SDF, DITTO) fit a parametric model to each observation at test time. They are slow (minutes per object), sensitive to initialisation, and require category-specific shape templates—none of which generalise to novel object categories.

Single-view feed-forward methods (Particulate, ArtSplat) amortise this cost by learning a category-level prior during training. Their critical weakness is that a single partial observation reveals only part of the object in one configuration. Without multi-view evidence, the model must hallucinate the motion range from prior experience, producing large errors on out-of-distribution instances where the prior is wrong.

Neither exploits the structure of multi-state observations. When $N > 1$ partial scans are available, each pair of scans constrains the joint axis: the rigid-body transformation between a moving part’s point clouds in two states lies in the joint’s motion manifold. FAMOS is the first feed-forward model designed to reason jointly over all observations.

Core Mechanism

Point encoding. Each point cloud X_n is encoded independently by a shared PointNet-style encoder ϕ into a feature map $F_n \in \mathbb{R}^{T \times D}$ ($D = 768$). $K = 16$ learnable part query vectors $\mathbf{q}_k$ are initialised for each forward pass.

Multi-state Articulation Transformer (MAT). The $L = 6$ transformer layers each alternate between two attention modes:

State-wise attention: For observation n , part queries cross-attend to point features of that observation only: $\mathbf{q}_k^{(n)} \leftarrow \text{Attn}(\mathbf{q}_k, F_n)$.

Global attention: All part queries from all observations attend jointly: $\{\mathbf{q}_k^{(n)}\}_{n,k} \leftarrow \text{Attn}(\{\mathbf{q}_k^{(n)}\}_{n,k}, \{\mathbf{q}_k^{(n)}\}_{n,k})$.

Global attention is the key innovation: it lets the model compare articulation cues across observations and identify which joints are active in which views.

Decoding. Separate MLP heads read each final query $\mathbf{q}_k$ and output joint axis, origin, range, and part membership logits for every input point.

Technical Formulation

The Observed Articulation Span (OAS) objective augments standard joint- parameter regression. Let $\theta_n^{(m)}$ be the joint angle of part m in observation n . The observed motion range is:

$$\Delta_m = \max_n \theta_n^{(m)} - \min_n \theta_n^{(m)}.$$

A regression head predicts $\hat{\Delta}_m$ from the part query $\mathbf{q}_k$:

$$\mathcal{L}_{\text{OAS}} = \sum_m \left(\hat{\Delta}_m - \Delta_m \right)^2.$$

The full loss combines part segmentation (Hungarian-matched cross-entropy), joint axis regression (cosine loss), origin regression (ℓ_2), motion range regression (ℓ_2), and OAS:

$$\mathcal{L} = \lambda_{\text{seg}} \mathcal{L}_{\text{seg}} + \lambda_{\text{axis}} \mathcal{L}_{\text{axis}} + \lambda_{\text{origin}} \mathcal{L}_{\text{origin}} + \lambda_{\text{range}} \mathcal{L}_{\text{range}} + \lambda_{\text{OAS}} \mathcal{L}_{\text{OAS}}.$$

Joint axis loss uses the unsigned cosine:

$$\mathcal{L}_{\text{axis}} = 1 - |\cos(\hat{\mathbf{a}}_m, \mathbf{a}_m)|.$$

Learning or Inference Procedure

1. Encode each observation independently with the shared PointNet encoder to obtain $\{F_n\}$.
2. Initialise K learnable part queries $\{\mathbf{q}_k\}$.
3. Run L MAT layers alternating state-wise and global attention.
4. Decode part queries via MLP heads into joint parameters and per-point part membership logits.
5. OAS head is used only during training; at inference, joint parameters are read from the main decoders.

Training data mixes real PartNet-Mobility instances with procedurally generated synthetic assets. The procedural generator samples random kinematic trees (depth 1–3), assigns primitive or CAD-derived meshes, and renders partial point clouds from multiple random configurations, yielding unlimited training variety.

What the Guarantee Says

FAMOS provides no formal approximation bound, but the paper demonstrates empirically that axis error drops monotonically as N increases from 1 to 4 observations on every benchmark. The largest single gain is from $N=1$ to $N=2$ (4.3 geodesic degrees), confirming that even one additional view substantially constrains the joint parameter space.

Experimental Findings

Three benchmarks cover in-distribution and out-of-distribution object types:

- **PartNet-Mobility:** 45 categories, 2,346 objects.
- **ACD:** Articulated CAD objects with novel geometries.
- **ArtiCraft-10K:** Large community-contributed dataset.

With 4 input observations (geodesic axis error in degrees, lower is better):

Method	PN-Mob	ACD	ArtiCraft
A-SDF	24.1	31.4	38.7
DITTO	21.3	28.6	35.2
Particulate	18.4	22.5	29.1
FAMOS	10.3	16.9	7.6

FAMOS’s biggest gain is on ArtiCraft (21.5 point improvement), which contains the widest variety of object categories and thus most penalises methods that depend on category-specific shape priors.

Ablations and Interpretation

Global attention is essential: Removing global attention and using only state-wise attention degrades axis error by 3.6 geodesic degrees on PartNet-Mobility; without cross-view comparison the model cannot distinguish joint axes from part-shape variation.

OAS objective matters: Removing OAS increases axis error by 2.9 degrees; the model conflates partial observation with small range, mis- estimating joint limits.

Synthetic data is the enabler: Removing procedural synthesis while keeping real data causes a 4.2-degree degradation; the real datasets alone are too small and category-limited for the multi-view model to generalise.

$N = 1$ is a competitive single-view baseline: Even with $N = 1$, FAMOS outperforms Particulate by 1.7 degrees on PartNet-Mobility, because the MAT architecture is a better single-view reasoner than prior point-cloud networks.

Reference

Kevin Qu, Stefan Ainsworth, Jiahao Wang, Gordon Wetstein, Qianqian Wang. **FAMOS: Feed-Forward 3D Articulation Modeling from Sparse Observations.** arXiv:2609.20817, September 2026. <https://arxiv.org/abs/2609.20817>